You are a Senior Software Architect, Technical Writer, and AI Solutions Consultant.

Analyze the entire repository and generate complete professional project documentation based only on the actual implementation.

Understand:
- Project objective
- Business problem solved
- End users
- Features
- System workflow
- Architecture
- Agent workflow
- LLM integration
- Data flow
- Technology stack
- Folder structure
- APIs
- Configuration files
- Deployment process

Generate documentation in the following format:

1. Executive Summary
- Project title
- Objective
- Problem statement
- Business need
- Solution overview
- Benefits
- Expected outcomes

2. Project Overview
- Introduction
- Scope
- Target users
- Use cases

3. Problem Statement
- Existing challenges
- Business impact
- Need for automation

4. Solution Architecture
- High-level architecture
- Component architecture
- System workflow
- Architecture explanation

5. Technology Stack
For each technology explain:
- What it is
- Why it is used
- Benefits

Cover:
- Frontend
- Backend
- Database
- AI/LLM
- APIs
- Libraries
- Deployment tools

6. Features
For each feature provide:
- Description
- Input
- Processing
- Output
- Business value

7. Agentic AI Architecture
Explain:
- All agents
- Responsibilities
- Inputs and outputs
- Agent interaction
- Workflow orchestration
- Prompt flow
- Result aggregation

8. LLM Integration
Explain:
- LLM provider
- Model used
- Prompt strategy
- Context flow
- Response generation
- Error handling

9. Data Flow
Explain complete execution flow:

User Input
→ Validation
→ Processing
→ Agent Execution
→ LLM Processing
→ Aggregation
→ Final Output

10. Module Description
For every major folder/module:
- Purpose
- Key files
- Responsibilities
- Dependencies

11. Workflow Explanation
Provide step-by-step execution from user action to final report generation.

12. API Documentation
If APIs exist:
- Endpoint
- Method
- Request
- Response
- Purpose

13. Database Documentation
If database exists:
- Schema
- Tables
- Relationships
- Data lifecycle

14. Installation Guide
- Prerequisites
- Setup steps
- Environment variables
- Running locally
- Production deployment

15. Testing Strategy
- Unit testing
- Integration testing
- Validation approach

16. Security Considerations
- Authentication
- Authorization
- Data protection
- Prompt safety
- API security

17. Performance & Scalability
- Current architecture limits
- Scaling opportunities
- Optimization strategies

18. Advantages
- Technical advantages
- Business advantages

19. Limitations
- Current constraints
- Known gaps

20. Future Enhancements
- Short-term improvements
- Long-term roadmap

21. Conclusion

Rules:
- Use actual code evidence only.
- Do not invent features.
- Explain in simple and technical language.
- Create professional company-grade documentation.
- Use diagrams/workflows in text format where helpful.
- Keep explanations concise but complete.
- Avoid repetition.
- Produce a final document ready for project submission and stakeholder review.